

Anirudh Chakravarty Kumar

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EDUCATION

University of Pennsylvania | GPA: 3.9/4.0

Philadelphia, US | Aug 2025 – May 2027

M.S.E. in Computer and Information Science

- Coursework: Trustworthy Machine Learning, Machine Learning, Big Data Analytics, Operating Systems

PES University | CGPA: 4.0/4.0

Bangalore, India | Dec 2020 – May 2024

B.Tech in Computer Science and Engineering

- Honors: Professor MRD Scholarship awarded 6 semesters in a row; Among top 20% of the batch

PUBLICATIONS

Exploring Syllable Discriminability during DDK Task across ALS Dysarthria Severity

INTERSPEECH 2024

Research Challenges and Gaps on the Recent Advances in Animal Emotion Recognition

ICITA 2024

WORK EXPERIENCE

Research Assistant | **Han AI Lab, Penn Medicine (UPenn)**

Jan 2026 – Feb 2026

- Developed self-supervised learning pipeline using VLMs on 400K+ unannotated chest X-ray images to detect latent pathology features

Software Engineer | **IDfy, Bangalore**

Jan 2024 – May 2025

- Built and scaled India's largest vehicle intelligence platform (10M+ records, 220M+ entries); co-founded VEHICLEAR, delivering vehicle legal/blacklist intelligence via production APIs (patent pending)
- Led LLM and applied ML initiatives, building OCR and line-detection pipelines that achieved 99.7% extraction accuracy across 220M+ records, addressing a core data extraction bottleneck that had persisted across prior systems for over 5 years

Research Intern | **Indian Institute of Science (IISc), Bangalore**

Jun 2023 – Aug 2023

- Conducted speech signal processing and deep learning research on syllable classification for ALS dysarthria analysis, including model development and human vs. model evaluation
- Co-authored INTERSPEECH 2024 paper: "Exploring Syllable Discriminability during Diadochokinetic Task with Increasing Dysarthria Severity for Patients with ALS"

PROJECTS

Sarize – AI Lecture Processing System | **University of Pennsylvania**

Apr 2026 – Apr 2026

- Built parallelized Whisper transcription + GPT-4o map-reduce summarization pipeline, converting a 90-min lecture to exam-ready material in under 10 minutes (9× reduction in processing time)
- Engineered multi-agent post-processing (file organization, SMTP delivery, 7-day spaced-repetition scheduling) for fully automated lecture-to-flashcard workflows

Differential Diagnosis Ranking via RL Fine-Tuning | **University of Pennsylvania**

Jan 2026 – May 2026

- Fine-tuned gpt-4.1-nano for 5-way differential diagnosis ranking across 6 medical specialties using SFT and DPO; haiku-teacher SFT achieved $\Delta\tau = +0.207$ and top-1 accuracy 59.0% vs. baseline 48.4%
- Showed self-distillation degrades ranking quality ($\Delta\tau = -0.066$) vs. a stronger external teacher; diagnosis normalization pipeline improved τ by +0.266 with zero model change

Multimodal Parkinson's Detection | **University of Pennsylvania**

Aug 2025 – Dec 2025

- Built multimodal deep learning models for Parkinson's detection by fusing DATScan SPECT and structural MRI, achieving near-state-of-the-art performance (AUROC = 0.98)
- Owned the DATScan baseline, including preprocessing, CNN modeling (ResNet, EfficientNet, Inception), and evaluation to guide fusion design

Penn OS | **University of Pennsylvania**

Aug 2025 – Dec 2025

- Designed preemptive multi-level scheduler with weighted CPU allocation (9:6:4), improving task fairness and reducing starvation to 0% under test workloads
- Built kernel-level process sleep, blocking, and wakeup mechanisms, ensuring starvation avoidance and correct scheduler behavior under concurrency

Multimodal Dog-Emotion Recognition | **PES University**

Aug 2023 – Aug 2024

- Developed a multimodal emotion recognition system for dogs using fusion-based ML classifiers over audio segments and key video frames, with separate image and audio pipelines
- Integrated object detection and generative AI for contextual inference and confidence scoring; resulted in an Indian Patent and an ICITA 2024 paper

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, MySQL, Node.js, React.js, Elasticsearch

Frameworks & Libraries: PyTorch, NumPy, scikit-learn, timm, FastAPI, Whisper, GPT-4o, CLIP, DINOv2, Nibabel, ffmpeg

AI/ML: VLMs, LLMs, Multimodal ML, Self-supervised Learning, Computer Vision, Medical Imaging, Speech Processing, CUDA/GPU Training, Agentic AI, OCR, Trustworthy ML

Tools & Platforms: Git, Docker, Kubernetes, SLURM, Google Cloud, Regular Expressions, Web Crawling, Data Cleaning